Sheet 1

Lecture problems

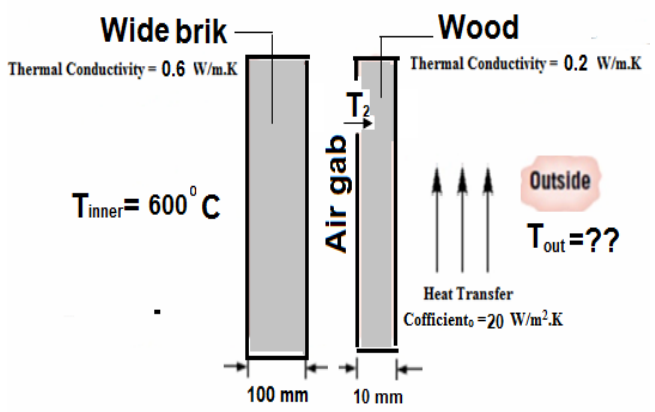
Ex1. A glass windowpane in a home is 0.620 cm thick and has dimensions of 1.00 m x 2.00 m. On a certain day, the temperature of the interior surface of the glass is 25.0°C and the exterior surface temperature is 0°C. (a) What is the rate at which energy is transferred by heat through the glass? (b) How much energy is transferred through the window in one day? [Thermal conductivity of glass 0.8W/m °C].

Ex 2. A wall consists of 25 cm wide brick and 15 cm wood separated by an air gap. The inner surface of the wall is at 1200°C and the room temperature is at 25°C. Calculate a) the rate at which heat is lost per m^2 of wall surface. The heat transfer coefficient from the outside wall surface to the air in the room is 20 W/m^2K, and the resistance to heat flow of the air gap is 0.16 m^2K/W. The thermal conductivity of the brick is 50 W/mK and the thermal conductivity of the wood is 10 W/mK. Also, b) calculate each interface temperature and the temperature of the outside surface of the wall.

Tutorial and Assignment Problems

Ex 3. A furnace wall is built in a room at a temperature of 20°C. The furnace wall consists of two layers of firebrick, each of width 180 mm and having a thermal conductivity of 0.9 W/mK, separated by an air gap. The outer layer is covered with a 25 mm layer of plaster whose thermal conductivity is 0.12 W/mK. The inner surface wall temperature is 1200°C and the heat transfer coefficient from the outside wall surface to the room is 20 W/m^2K and the resistance of the air gap is 0.2 K/W. Calculate the interface temperatures.

Ex 4. A wall consists of 100 mm wide brick and 10 mm wood separated by an air gab as shown in figure. The thermal conductivity of the wide brick and wood equal 0.6 W/mK and 0.2 W/mK, respectively and the heat transferee coefficient from the out side wall surface is 20 W/m^2K. If the resistance of the air gab 0.2 K/W and the inner surface temperature of the wall is at 600 °C and the second interface temperature equal T_2= 160 °C (see the figure). Find the rate of heat transferred per unit area

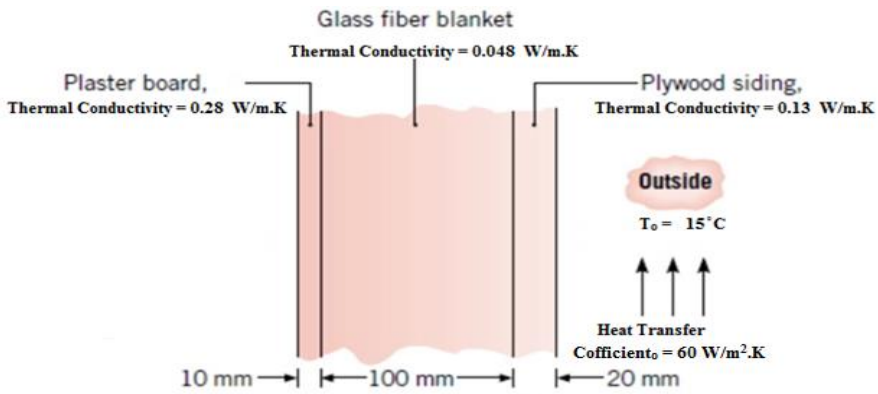


Ex 5. The surface of the Sun has a temperature of about5 800 K. The radius of the Sun is 6.96 x10^8 m. Calculate the total energy radiated by the Sun each min. Assume the emissivity of the Sun is 0.986 and the Stefan constant **5.67 x 10^{-8} W/m²K⁴**.

Ex 6. A student is trying to decide what to wear. His bedroom is at 20.0°C. His skin temperature is 35.0°C. The area of his exposed skin is 1.50 m^2. People all over the world have skin that is dark in the infrared, with emissivity about 0.900. Find the net energy transfer from his body by radiation in 10.0 min (the Stefan constant **5.67 x 10^{-8} W/m²K⁴**).

Ex 7. A furnace wall consists of 20 cm firebrick, 10 cm insulating brick and 15 cm building brick, respectively. The inside wall is a temperature of 800oC and the atmospheric temperature is 15oC. Calculate the heat loss per m2 of wall area and the temperatures of the interfaces of the furnace. The heat transfer coefficient for the outside surface is 8 W/m2K and the thermal conductivities of the firebrick, insulating brick, and building brick are 4, 0.3, and 0.8 W/mK, respectively.

Ex 8. Calculate the interfaces temperatures for the composite wall is consists of 3 layer with their properties as shown in the figure. The inside wall surface temperature is 750ºC.



Sheet 2

Lecture problems

Ex1. A certain fluid at $\mathbf{10}\ \boldsymbol{bar}$ is contained in a cylinder behind a piston, the initial volume being $\mathbf{0.05}\ \boldsymbol{m^3}$. Calculate the work done by the fluid when it expands at constant pressure to a final volume of $\mathbf{0.2}\ \boldsymbol{m^3}$.

Ex2 . 1 kg of a fluid at a pressure of $3\ bar$, and with a specific volume of $0.18\ m^3/Kg$, contained in a cylinder behind a piston expands reversibly to a pressure of $0.6\ bar$ according to a law $pv^2 = c$, where c is a constant. Calculate the work done during the process and sketch the P-V diagram.

Tutorial Problems and Assignments

Ex 3. 500 gm of fluid has an internal energy of $10\ kJ$ is heating from 20 ^oC to 80 ^oC, until its internal energy is $20\ kJ$. If the specific heat capacity of the fluid 4100 J/ kg ^oC, Calculate: a- The Heat flow. b- The work done and **whether it is on or by the fluid.**

Ex 4. Ex.1 kg bar of copper is heated isobarically at atmospheric pressure (1 x10^2 K Pa). If its temperature increases from 20°C to 50°C (Taking the linear expansion coefficient of copper 1.7x10^{-5} $^0c^{-1}$, density of the copper 8.92x10^3 kg/m^3, and the specific heat capacity of the copper 387J/kg. 0^C). Find

(A) The work done?
(B) The heat energy that transferred to the copper bar?
(C) The increase in internal energy of the copper bar?

Ex 5. Ex. 400 gm of water at 100°C vaporizes isobarically at pressure 10 bar from 0.02 m^3 to 0.06m^3 . If the latent heat of vaporization equal 2.26 x 10^6J/kg, find

A- The work done and **whether it is on or by the gas**
B- The change in internal energy and **whether it is a gain or a loss.**

Ex 6. 200 gm of gas is compressed at constant pressure 10 bar from 0.08 m^3 to 0.04 m^3 . If the internal energy of the fluid is changed by 5000 J, find

I- The work done and **whether it is on or by the gas**
II- The Heat flow and **whether it is absorb or rejected from the system.**
III- The change in temperature of the gas, if the specific heat capacity of gas is 1000J/Kg^{0}C.

Sheet 3

Lecture problems

Ex1. 0.8 kg of carbon dioxide, CO_2, at 20 bar and stored in a vessel of 0.04 m^3. If its molecular mass equal 44, and the value of γ equal 1.3. Find a) the specific gas constant, R, b) the specific heat capacity at constant pressure C_P , c) the specific heat capacity at constant volume, C_v, d) the internal energy , and e) the enthalpy.

Ex2. A rigid tank contains 1.5 moles of an ideal gas. Determine the number of moles of gas that must be withdrawn from the tank to lower the pressure of the gas from 25 atm to 5 atm. Assume the volume of the tank and the temperature of the gas remain constant during this operation.

Tutorial Problems and Assignments

Ex3. A gas is confined in a tank at a pressure of 11.0 atm and a temperature of $25.0^o C$. If two-thirds of the gas is withdrawn and the temperature is raised to $75.0^o\ C$, what is the pressure of the gas remaining in the tank?

Ex 4. A certain perfect gas of mass $0.01\ kg$ occupies a volume of $0.003\ m^3$ at a pressure of $7\ bar$ and a temperature of $131°C$. The gas is allowed to expand until the pressure is $1\ bar$ and the final volume is $0.02\ m^3$. Calculate:

i. The molar mass of the gas.
ii. The final temperature.

Ex 5. Oxygen, O_2, at 200 bar is to be stored in a steel vessel at 20 ℃. The capacity of the vessel is 0.04 m^3. Assuming that O_2 is a perfect gas, calculate the mass of oxygen that can be stored in the vessel. The vessel is protected against excessive pressure by a fusible plug which will melt if the temperature rises too high. At what temperature must the plug melt to limit the pressure in the vessel to 240 bar? The molar mass of oxygen is 32 kg/kmol.

Ex 6. The relative molecular mass of carbon dioxide, CO_2, is 44. In an experiment the value of γ for CO_2 was found to be 1.3. Assuming that CO_2 is a perfect gas, calculate the specific gas constant, R, and the specific heat capacities at constant pressure and constant volume, $C_{P\ and}\ C_v$.

Ex 7. When a certain perfect gas is heated at constant pressure from 15 ℃ to 95 ℃, the heat required is 11.36 kJ/kg. When the same gas is heated at constant volume between the same temperatures the heat required is 808 kJ/kg. Calculate c_p, c_v, γ, R and the molar mass of the gas.

Ex 8. A vessel of volume $0.2\ m^3$ contains Nitrogen at $1.013\ bar$ and $15°C$. If $0.2\ kg$ of Nitrogen is now pumped into the vessel, calculate the new pressure when the vessel has returned to its initial temperature. The molar mass of nitrogen is $28\ kg/kmol$, and it may be assumed to be a perfect gas.

Sheet 4

Lecture problems

Ex1. 1 kg of air at 1.02 bar, 20 °C is compressed reversibly according to a law $pv^{1.3}$= constant, to a pressure of 5.5 bar. Calculate the work done on the air and the heat supplied during the compression (molecular mass = 29 kg/kmole, γ = 1.4).

Ex2. One kg of a perfect gas (molar mass M=29 kg/kmole and γ = 1.3) is compressed adiabatically from a pressure of 40 bar and a temperature of 77 ^{o}C to a pressure of 130 bar. The gas is then expanded isothermally and finally, it is cooled at a constant pressure back to its initial state. Calculate the **net work done**, **the net heat flow** and the **net change in entropy** for the cycle. **Sketch** the cycle on both **p-v diagram, and T-S diagram**.

Ex3. A certain perfect gas at a pressure of 10 bar and a temperature of 97 ^{o}C occupies a volume of 0.04 m^3. The gas is allowed to expand adiabatically to a pressure of 4.5 bar then, it is heated at a constant pressure. Finally, it is compressed isothermally back to its initial conditions. Calculate **the net work done** and **the net heat flow** and the **net change in entropy** for the cycle. **Sketch** the cycle on both **p-v diagram, and T-S diagram**. (molar mass is 32 kg/kmole,c_v=0.65 kJ/kg.K).

Tutorial Problems and Assignments

Ex 4. Oxygen (molar mass 32 kg/kmol) expands reversibly in a cylinder behind a piston at a constant pressure of 3 bar. The volume initially is 0.01 m^3 and finally is 0.03 m^3; the initial temperature is 17 °C. Calculate the work input and the heat supplied during the expansion. Assume oxygen to be a perfect gas and take c_p=0.917 kJ/kg K.

Ex 5. A certain perfect gas of mass $2\ kg$ occupies a volume of $0.4\ m^3$ at a pressure of 10 bar. The gas is heating at constant volume until the pressure is $20\ bar$. If the molecular mass of the gas equal 32, and Cv =0.647, Calculate:
a- The specific gas constant if. b- The work done and the heat flow.

Ex6. Nitrogen (molar mass 28 kg/ kmol) expands reversibly in a perfectly thermally insulated cylinder from 3.5 bar, 200 °C to a volume of 0.09 m^3. If the initial volume occupied was 0.03 m^3, calculate the work input. Assume nitrogen to be a perfect gas and take c_v=0.741 kJ/kg K.

Ex7. A certain perfect gas is compressed reversibly from 1 bar 17 °C to a pressure of 5 bar in a perfectly thermally insulated cylinder, the final temperature being 77 °C. The work done on the gas during the compression is 45 kJ/kg. Calculateγ, c_v, R, and the molar mass of the gas.

Ex 8. 1 Kg of Nitrogen (molecular mass = 28 kg/kmole, γ = 1.4) is compressed adiabatically and reversibly in a cylinder behind a piston from a pressure of 10 bar and 127 ^{o}C to a pressure of 60 bar. The gas is then expanded according to a law $pv^{1.3}$ = constant. It is finally cooled at a constant pressure back to its original condition. Calculate the net work done and the net heat flow for the cycle and the **net change in entropy** for the cycle. **Sketch** the cycle on both **p-v diagram, and T-S diagram**.

Ex 9. 0.14 kg of air at 1.2 bar and 0.09 m^3 are compressed polytropically to volume 0.03 m^3 in such a way that one half of the work input is rejected as heat. It is then heated at constant volume until the temperature is 140 ^{0}C. Finally, it expands adiabatically back its initial conditions. Calculate the net work done and the net heat flow and the **net change in entropy** for the cycle. **Sketch** the cycle on both **p-v diagram, and T-S diagram**.

Ex 10. One kg of air at a pressure of 15 bar and temperature 127 ^{o}C is allowed to expand according to the law $pv^{1.2}$ = c until the pressure becomes 10 bar. The air is then heated at constant volume. Finally, it is compressed back to its initial conditions according to a law pv = c. Calculate the work done, the heat flow and Sketch the cycle p-v diagram.

Sheet 5

Lecture problems

Ex 1. A particular heat engine has a mechanical power output of 5.00 kW and an efficiency of 25.0%. The engine expels 8.00×10^3 J of exhaust energy in each cycle. Find (a) the energy taken in during each cycle and (b) the work done for each cycle. (c) The time interval for each cycle

Ex 2. A refrigerator has a coefficient of performance equal to 5.00. The refrigerator takes in 120 J of energy from a cold reservoir in each cycle. Find: a) The work required in each cycle b) The energy expelled to the hot reservoir.

Ex 3. What is the coefficient of performance of a refrigerator that operates with Carnot efficiency between temperatures 23.00°C and 127.0°C?

Tutorial and Assignment Problems

Ex 4. The work done by an engine equals one-fourth the energy it absorbs from a reservoir.

i- What is its thermal efficiency?
ii- What fraction of the energy absorbed is expelled to the cold reservoir?

Ex 5. A heat engine takes in 360 J of energy from a hot reservoir and performs 25.0 J of work in each cycle. Find:

I- The efficiency of the engine
II- The energy expelled to the cold reservoir in each cycle.

Ex 6. During each cycle, a refrigerator ejects 625 kJ of energy to a high-temperature reservoir and takes in 550 kJ of energy from a low-temperature reservoir. Determine:

a- The work done on the refrigerant in each cycle and
b- The coefficient of performance of the refrigerator.

Ex 7. A heat engine operates between a reservoir at 25.0°C and one at 375°C. What is the maximum efficiency possible for this engine?

Ex 8. One of the most efficient heat engines ever built is a coal-fired steam turbine in the Ohio River valley, operating between 1 870°C and 430°C.

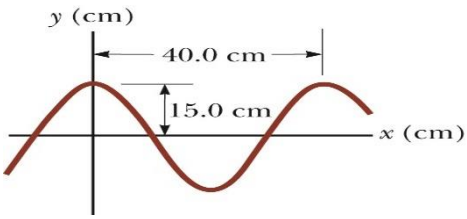
I-What is its maximum theoretical efficiency?

II- The actual efficiency of the engine is 42.0%. How much mechanical power does the engine deliver if it absorbs 1.40 3 105 J of energy each second from its hot reservoir?

Sheet 6

Lecture problems

Ex1. A wave traveling in the positive x direction as shown in figure. Find the amplitude, wavelength, periodic time, and speed of wave if it has a frequency of 8.0 Hz.



Ex2. A traveling wave is described by the equation y(x,t) = (0.003) cos (20 x + 200 t), where y and x are measured in meters and t in seconds. What are the amplitude, frequency, wavelength, speed and direction of travel for this wave?

Ex3. A harmonic wave on a string has amplitude of 30 cm, a wavelength of 2 m, and a velocity of 12 m/s in the positive x direction. At time t=0, this wave has a crest at x=0.

A. Formulate the mathematical equation describing the general wave describing the displacement of an element of a string (x,t).
B. Find the phase shifting.
C. Determine the frequency, the angular frequency, and the wave number of this wave.
D. Find the displacement at x= 0.2 m and t=2 s.

Tutorial and Assignment Problems

Ex 4. Choose the Correct Answer

a. Which of the following quantities is transferred during wave propagation?
   i. Energy ii. Speed iii. Matter

b. If you double the frequency of a vibrating object, what happens to the period?
   i. doubled ii. Stay constant iii. Fall to the half

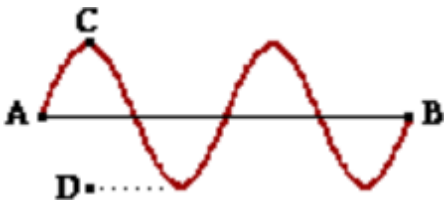
c. Mechanical waves could be transferred through
   i. Space ii. solid, liquid, gas iii. both i and ii

d. The wave speed is …. …to the angular frequency
   i. inversely proportional ii. directly proportional iii. nonrelative

e. Which of the following isn't an example of electromagnetic waves?
   i. radio-waves ii. ultrasound waves iii. microwaves

Ex 5. The distance between two successive crests is 1.2 m. Eight crests pass through a given point each 12 seconds. Calculate the wave speed, wavelength, periodic time, and speed of this wave.

Ex6. A wave with a frequency of 12.3 Hz is traveling from left to right across a rope as shown in the diagram at the right Positions A and B in the diagram are separated by a horizontal distance of 42.8 cm. Positions C and D in the diagram are separated by a vertical distance of 12.4 cm. Determine the amplitude, wavelength, period, speed of this wave.



Ex7. The wave function on a traveling wave on a taught spring is given by the following equation:

$$y(x,t)=0.35\sin(10\pi t+2\pi x+\pi/2\ )$$

A. What is the speed and the direction of the wave travel?
B. What is the displacement of an element locating at t=0 and x=0.1m
C. The wavelength, the frequency and the period of this wave

Ex 8. A sinusoidal wave is described by the wave function

$$y=0.25\ \sin\ (0.3\ x-40t)$$

- **Determine** the amplitude, the angular frequency, the wave number, the wavelength, the wave speed and the direction of motion. Also calculate the displacement of the wave at x=2 m and t=0.3 s.

Ex 9. A harmonic wave on a string has amplitude of 2.0 cm, a wavelength of 1.2 m, and a velocity of 6.0 m/s in the positive x direction. If the displacement of the wave equal zero at time t=0, and x=0.

E. Formulate the mathematical equation describing the general wave describing the displacement of an element of a string (x,t).
F. Determine the frequency, the angular frequency, and the wave number of this wave.
G. Find the displacement at x= 0.2 m and t=2 s

Sheet 7

Lecture problems

Ex1. Find the speed of sound through air at 0 ^oC, given that the Bulk modulus of air is 1.414 x 10^5 $\frac{N}{m^2}$ and its density is 1.29 $\frac{kg}{m^3}$. Then find the speed of sound through air at 30^oC.

Ex2. A fish boat sends sound wave signal in air with a power output of 100W. Assume the source is point source,

a- Find the received **intensity** by another boat at distance of 2 m, and the **sound level**.

b- Find the distance at which the received intensity by the other boat is reduced to a level of 120 dB. **[Io= 10^{-12} W/m]**

Tutorial and Assignment Problems

Ex3. If the density of aluminium is 2.7× 10^3 kg/m^3 and that the speed of sound in aluminium is 5100 m/s, calculate Young's modulus for this material.

Ex4. A sound wave propagates in air at 27 ^oC with frequency 4 kHz. It passes through a region where the temperature gradually changes and then moves through air at 0 ^oC with velocity 331m/s. Explain what happen to the speed of sound wave? the frequency and the wavelength?

Ex5. The velocity of sound in sea water is $1.53 * 10^3 \frac{m}{s}$. Find the bulk modulus (in $\frac{N}{m^2}$) of sea water if its density is $1.03 * 10^3 \frac{kg}{m^3}$.

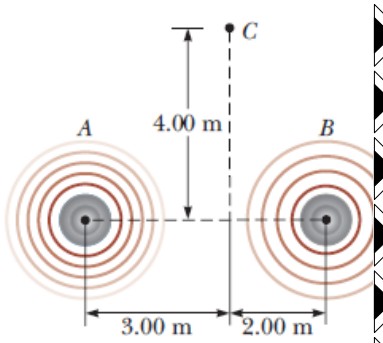
Ex6. A source emits sound waves with a power output of 80 W. Assume the source is a point source. Find:

A. The intensity at a distance 3 m from the source.
B. Find the distance at which the sound reduces to a level of 40 dB.

Ex7. Two small speakers emit sound waves of different frequency equally in all directions. Speaker A has an output of 1 mW and speaker B has an output of 1.5 mW.

Determine the sound level in DB at point C in each of the following cases:

A. Only Speaker A is emitting the sound.
B. Only speaker B is emitting the sound.
C. Both speakers emit the sound.



Ex8. A source of sound (1000 Hz) emits uniformly in all directions. An observer 3 m from the source measures a sound level of 40 dB. Calculate the average power output of the source. [Hint: I, = 10^{-12} W/m²].

Sheet 8

Lecture problems

Ex1. Write down the equation of the frequency (f') heard by a stationary observer if the Source is moving away from the observer?

Ex2. Write down the equation of the frequency (f') heard by an observer follow the source with the same direction?

Ex3. A submarine (sub A) travels through water at a speed of 8.00 m/s, emitting a sonar wave at a frequency of 1400 Hz. The speed of sound in the water is 1533 m/s. A second submarine (sub B) is located such that both submarines are traveling directly toward each other. The second submarine is moving at 9.00 m/s.

a) What frequency is detected by an observer riding on sub B as the subs approach each other?

b) The subs barely miss each other and pass. What frequency is detected by an observer riding on sub B as the subs recede from each other?

Tutorial and Assignment Problems

Ex 4. Suppose a police car is at rest and emits at 1200 Hz. What frequency would a person hear if he is moving at 30 m/s (a) toward it, (b) away from it? Take the speed of sound in air 341 m/s.

Ex5. A train is moving parallel to a highway with a constant speed of (20 m/s). A car is travelling in the same direction as the train with a speed of (40 m/s). The car horn sounds at a frequency of (510 Hz). And the train whistles sounds at a frequency of (320 Hz). [Hint: For air; speed of sound= 343 m/s]

A. When the car is behind the train what frequency does a man in the car observe for the train whistle?

B. When the car is in front of the train what frequency does a train passenger observe for the car horn?

Ex6. Two engines pass each other moving in opposite directions with uniform speed of 30m/s. One of them is blowing a whistle of frequency 540Hz. Calculate the frequency heard by driver of second engine before they pass each other. Speed of sound is 330m/sec.

Ex7. The security alarm on a parked car goes off and produces a frequency of 960 Hz. The speed of sound is 343 m/s. As you drive toward this parked car, pass it, and drive away, you observe the frequency of 900 Hz. At what speed are you driving?